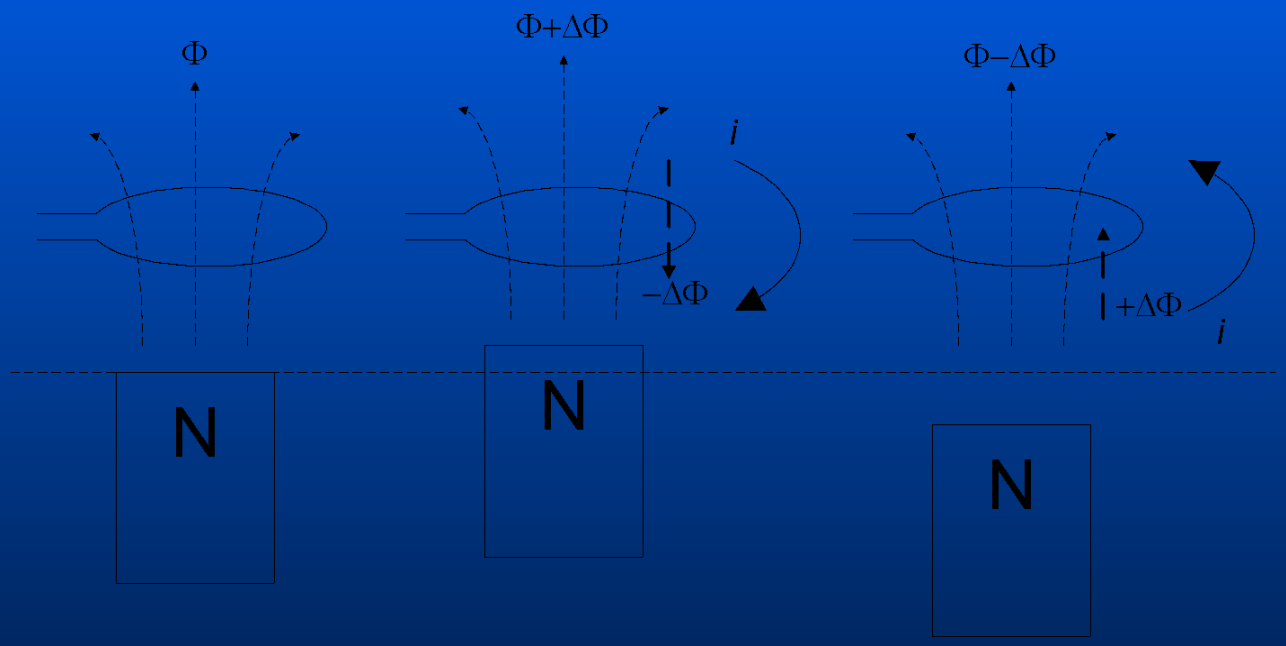
제 7 장 직류전원장치

DC Power Supplies

7.1 개요

- 직류-직류 변환기의 단점
  - 입력단과 출력단이 전기적으로 결합
  - 동일 접지를 사용하여야함
- 변압기 사용시의 장점
  - 입출력단이 분리되는 문제점 해소
  - 고주파 변압기 사용: 크기 및 가격 저렴
- 변압기의 입력과 출력 해석

7.2 변압기 모델: 전자유도

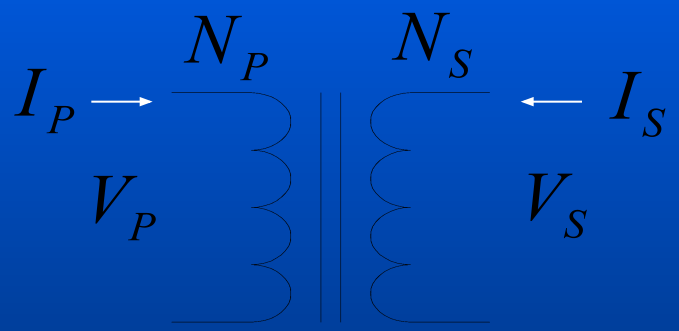


- Faraday's Law(전자유도법칙)

$$emf = -\frac{d\phi}{dt}$$

emf: electromotive force

7.2 변압기 모델: 전압, 전류



1차 (Primary)

2차(Secondary)

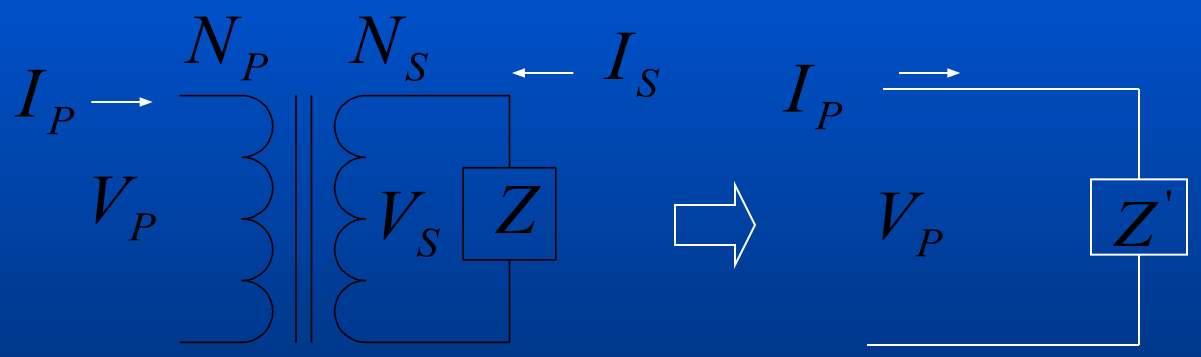
- 전압, 전류

$$V_P = N_P \frac{d\phi}{dt},\ V_S = N_S \frac{d\phi}{dt},\ \therefore \frac{V_p}{V_S} = \frac{N_p \frac{d\phi}{dt}}{N_S \frac{d\phi}{dt}} = \frac{N_p}{N_S}$$

- 권수비

$$\frac{N_p}{N_s} = \frac{V_P}{V_S} = \frac{I_S}{I_P} = a \qquad \because\ V_P I_P = V_S I_S$$

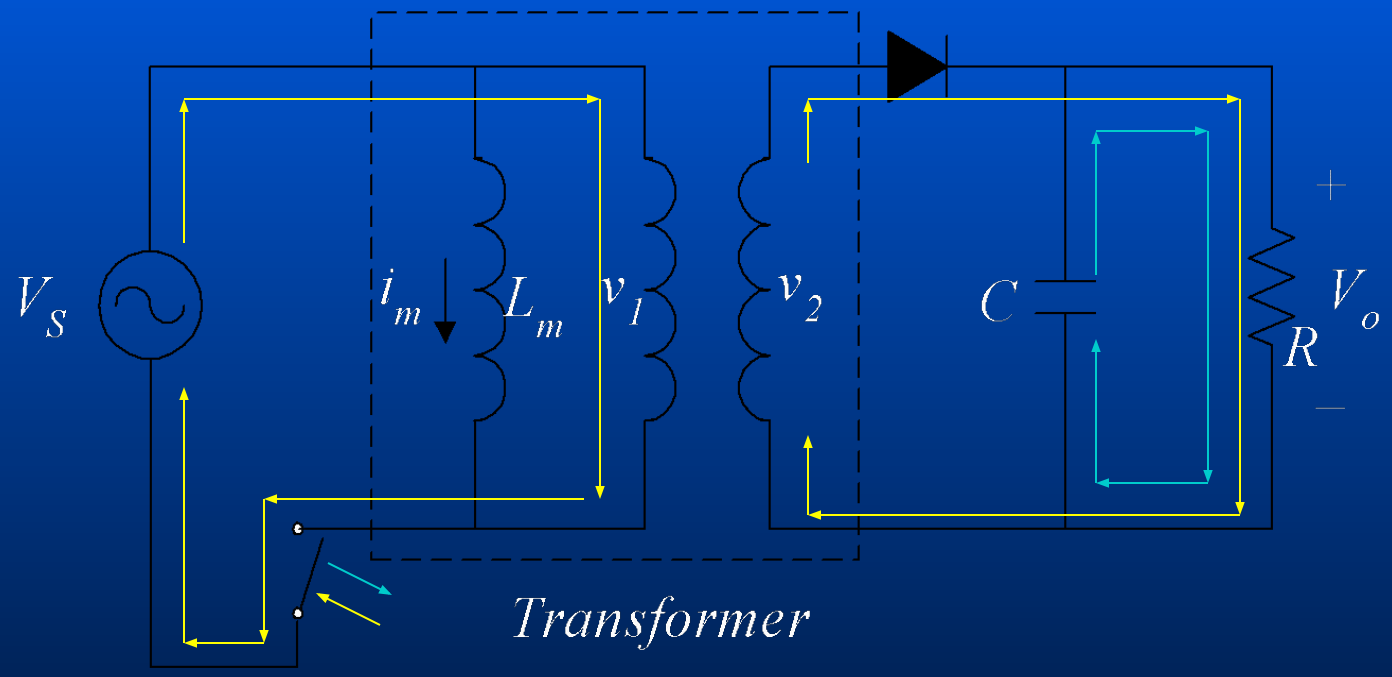
7.2 변압기 모델: 임피던스 변환



- 임피던스 변환(2차→1차)

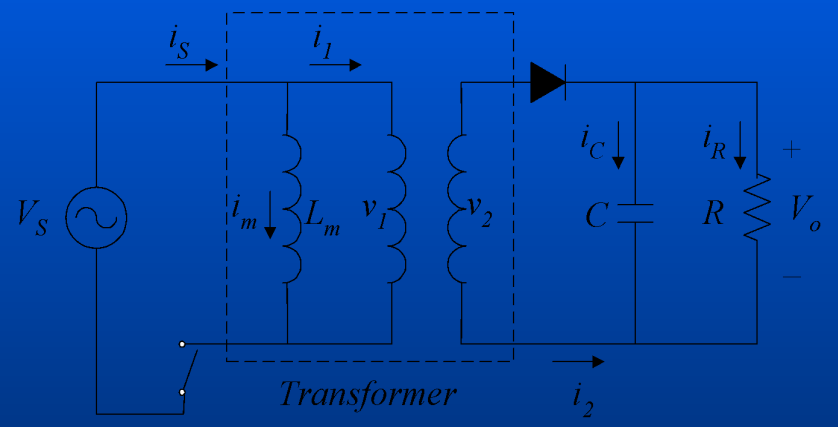
$$Z' = \frac{V_P}{I_P} = \frac{aV_s}{\frac{I_S}{a}} = \frac{a^2 V_s}{I_S} = a^2 Z$$

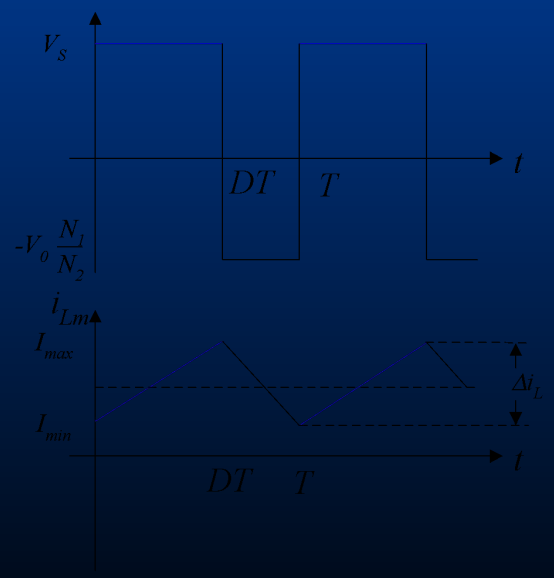
7.3 플라이백 변환기(Flyback converter)



- Switch on: 변압기에 의해 R에 전류/동시에 C 충전
- Switch off: C 충전된 전하가 방전 파형이 평활

7.3 플라이백 변환기: 전류





- Switch On

$$v_1 = V_S = L_m \frac{di_{Lm}}{dt}$$

$$\frac{\Delta i_{Lm}}{\Delta t} = \frac{\Delta i_{Lm}}{DT} = \frac{V_S}{L_m}$$

$$(\Delta i_{Lm})_{closed} = \frac{V_S DT}{L_m}$$

- Switch Off

$$\boxed{v_2 - V_0 = 0}$$

$$v_1 = v_2 \frac{N_1}{N_2} = -V_0 \frac{N_1}{N_2} = L_m \frac{di_{Lm}}{dt}$$

$$\frac{\Delta i_{Lm}}{\Delta t} = \frac{\Delta i_{Lm}}{(1-D)T} = \frac{-V_0}{L_m} \frac{N_1}{N_2}$$

$$(\Delta i_{Lm})_{open} = \frac{-V_0 (1-D)T}{L_m} \frac{N_1}{N_2}$$

7.3 플라이백 변환기: 전류

- Switch On

$$(\Delta i_{Lm})_{closed} = \frac{V_S DT}{L_m}$$

- Switch Off

$$(\Delta i_{Lm})_{open} = \frac{-V_0(1-D)T}{L_m}\frac{N_1}{N_2}$$

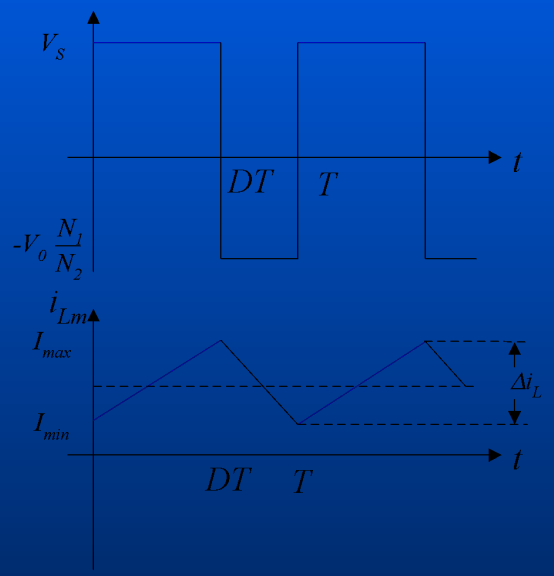
- 충전전류+방전전류=0

$$\frac{V_S DT}{L_m} - \frac{V_0(1-D)T}{L_m}\frac{N_1}{N_2} = 0$$

$$\frac{V_S DT}{L_m} = \frac{V_0(1-D)T}{L_m}\frac{N_1}{N_2}$$

$$\therefore V_0 = V_S \frac{D}{1-D}\frac{N_2}{N_1}$$

7.3 플라이백 변환기: 인덕터전류



- 인덕터의 평균전류

$$V_S I_{Lm} D = \frac{V_0^2}{R} = \frac{\left(V_S \frac{D}{1-D} \frac{N_2}{N_1}\right)^2}{R} = \frac{D^2 V_S^2}{(1-D)^2 R}\left(\frac{N_2}{N_1}\right)^2$$

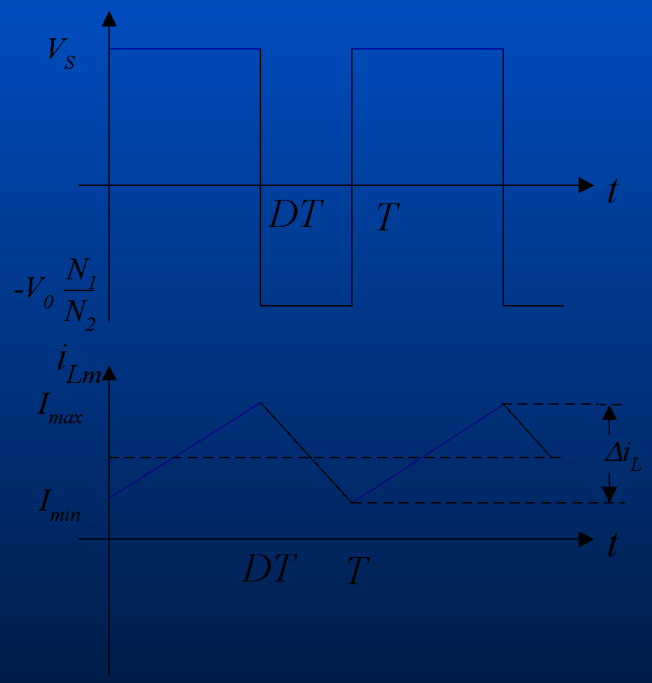
$$\therefore I_{Lm} = \frac{V_S D}{(1-D)^2 R}\left(\frac{N_2}{N_1}\right)^2 = \frac{V_0}{(1-D)^2 R}\left(\frac{N_2}{N_1}\right)^2$$

- 인덕터의 전류 최소, 최대값

$$I_{Lm,\max} = I_{Lm} + \frac{\Delta i_{Lm}}{2} = \frac{V_S D}{(1-D)^2 R}\left(\frac{N_2}{N_1}\right)^2 + \frac{V_S DT}{2L_m} \quad \left(\because \frac{\Delta i_{Lm}}{\Delta t} = \frac{\Delta i_L}{DT} = \frac{V_S}{L_m}\right)$$

$$I_{Lm,\min} = I_{Lm} - \frac{\Delta i_{Lm}}{2} = \frac{V_S D}{(1-D)^2 R}\left(\frac{N_2}{N_1}\right)^2 - \frac{V_S DT}{2L_m}$$

7.3 플라이백 변환기: 스위칭 주파수



- 전류 연속: 0 보다 커야함

$$I_{L\min} = 0 = \frac{V_S D}{(1-D)^2 R}\left(\frac{N_2}{N_1}\right)^2 - \frac{V_S DT}{2L_m}$$

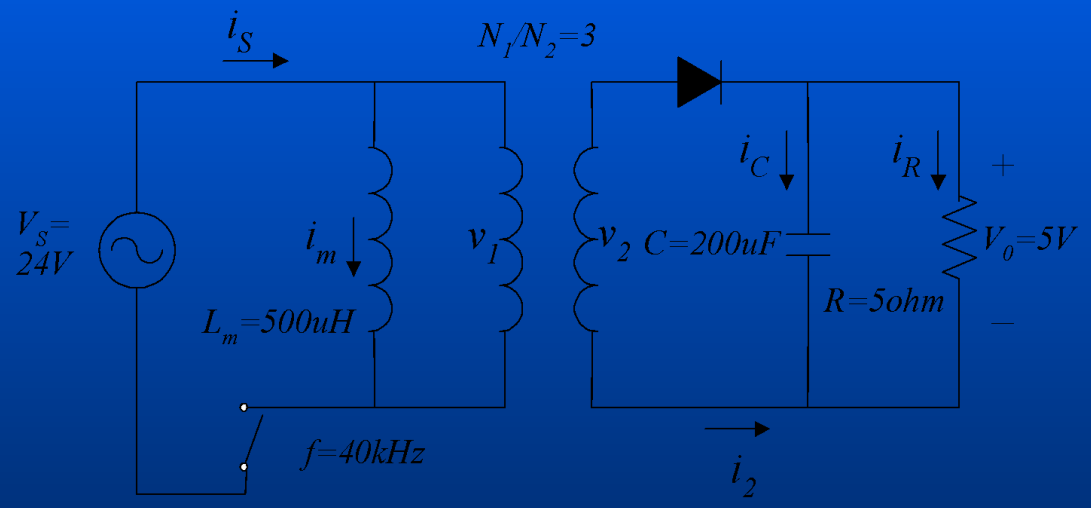
$$I_{L\min} \frac{V_S D}{(1-D)^2 R}\left(\frac{N_2}{N_1}\right)^2 = \frac{V_S DT}{2L_m} = \frac{V_S D}{2L_m f}$$

- 인덕터 및 스위치의 조합

$$(Lf)_{\min} = \frac{(1-D)^2 R}{2}\left(\frac{N_2}{N_1}\right)^2$$

$$\therefore L_{\min} = \frac{(1-D)^2 R}{2f}\left(\frac{N_2}{N_1}\right)^2$$

예제 7-1 플라이백 변환기(1)

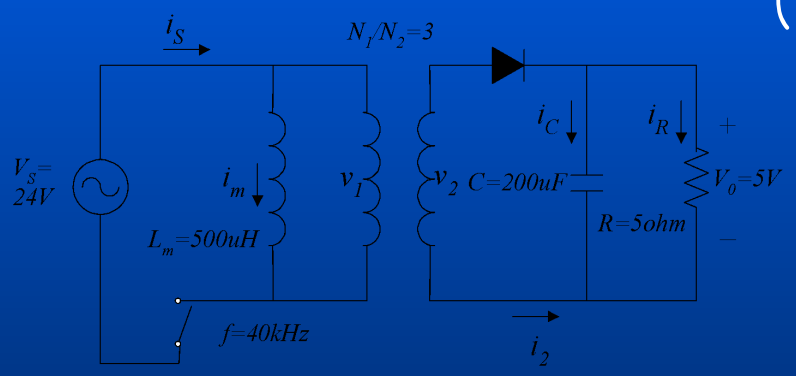


(a) 필요한 듀티비

$$V_0 = 5 = V_S \frac{D}{1-D}\frac{N_2}{N_1} = 24\left(\frac{D}{1-D}\right)\frac{1}{3}$$

$$D = \frac{5}{8}(1-D), \quad \frac{13}{8}D = \frac{5}{8}, \quad \therefore D = \frac{5}{13} = 0.385$$

예제 7-1 플라이백 변환기(2)



(b) 평균전류

$$\therefore I_{Lm} = \frac{V_0^2}{V_S DR} = \frac{5^2}{(24)(0.385)(5)} = 541mA$$

$$\Delta i_{Lm} = \frac{V_S D}{L_m f} = \frac{(24)(0.385)}{500(10)^{-6}(40000)} = 482mA$$

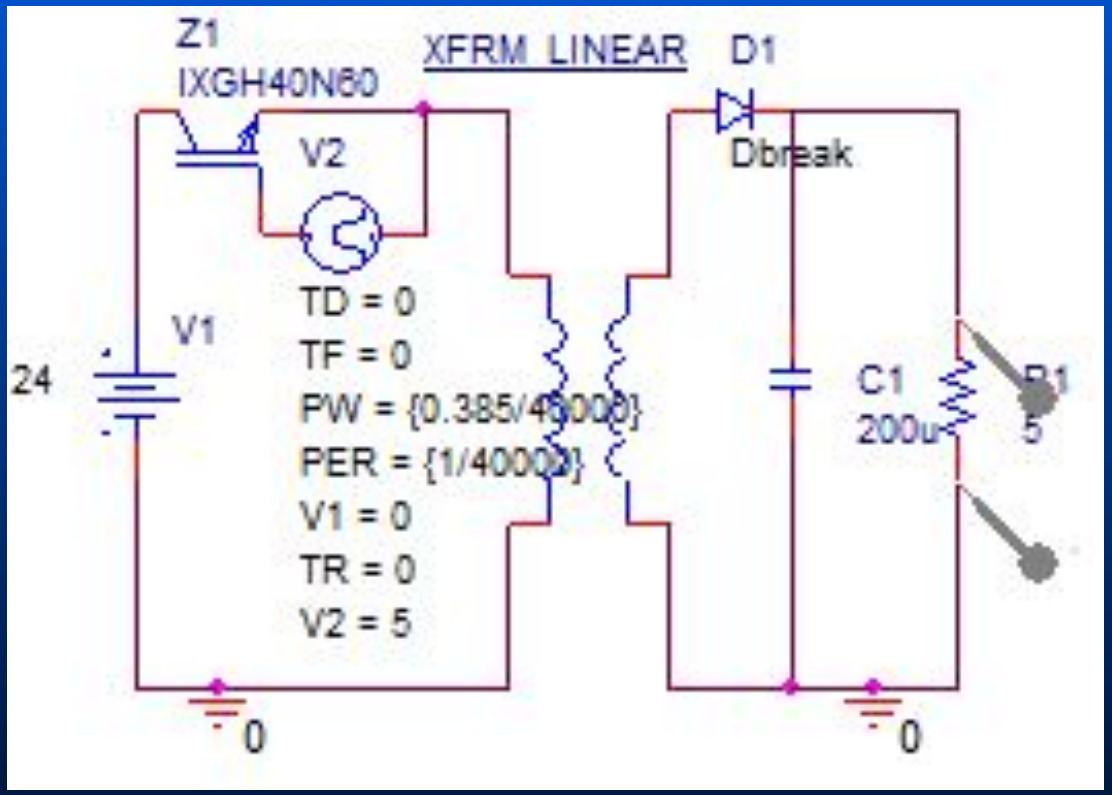
$$I_{Lm,\max} = I_{Lm} + \frac{\Delta i_{Lm}}{2} = 541 + \frac{462}{2} = 772mA$$

$$I_{Lm,\min} = I_{Lm} - \frac{\Delta i_{Lm}}{2} = 541 - \frac{462}{2} = 310mA$$

(c) 맥동

$$\frac{\Delta V_0}{V_0} = \frac{D}{RCf} = \frac{0.385}{(5)200(10)^{-6}(40000)} = 0.96\%$$

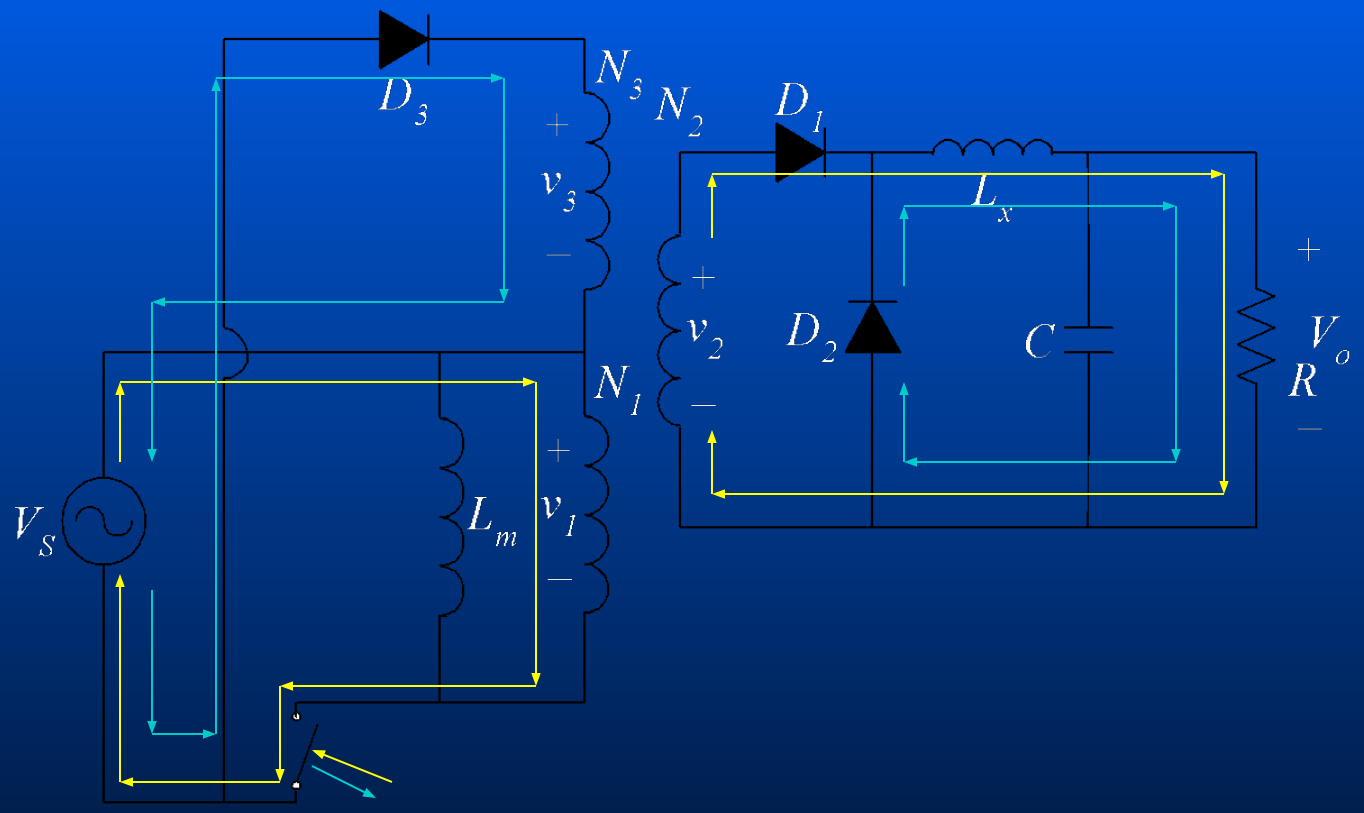
예제 7-1 플라이백 변환기(OrCAD 해)



- *K3019PL_3C8*
- *XFRM_LINER*

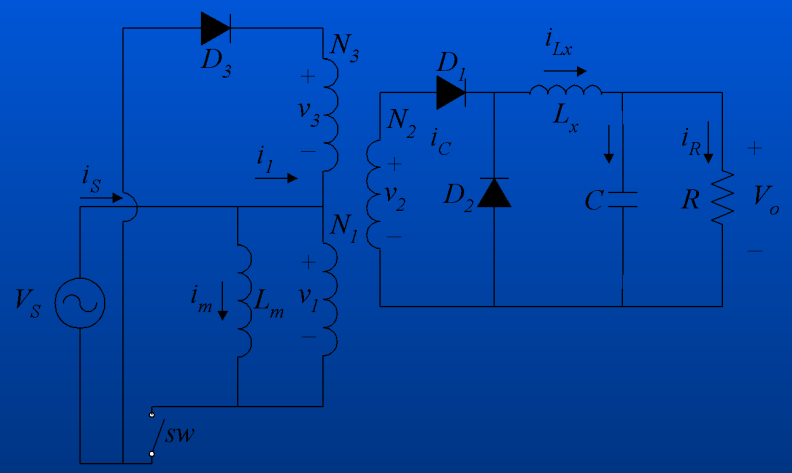
- *Parameter*
  - ✔ V_S=24 V
  - ✔ R=5 ohm
  - ✔ C=200 μF
  - ✔ TX1
    - Coupling : 0.99
    - L1_turns : 30
    - L1_turns : 10
  - ✔ f=25 kHz(T=40 μS)
  - ✔ D=0.6 (D=24 μS)
  - ✔ Step : 0 : 1u: 10 ms
- *Find : 출력전압*

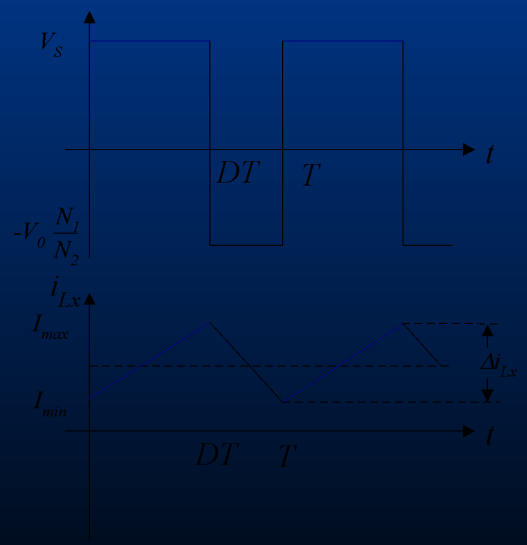
7.4 포워드 변환기(Forward converter)



- Switch on: 변압기에 의해 R에 전류/동시에 LC 충전
- Switch off: 환류 다이오드를 통해 방전/평활

7.4 포워드 변환기: 전류





- Switch On

$$v_{Lx} = v_2 - V_0 = V_S\left(\frac{N_2}{N_1}\right) - V_0 = L_x \frac{di_{Lx}}{dt}$$

$$\frac{\Delta i_{Lx}}{\Delta t} = \frac{\Delta i_{Lm}}{DT} = \frac{V_S\left(\frac{N_2}{N_1}\right) - V_0}{L_x}$$

$$(\Delta i_{Lx})_{closed} = \left[V_S\left(\frac{N_2}{N_1}\right) - V_0\right]\frac{DT}{L_x}$$

- Switch Off

$$v_3 = -V_S, \; v_{Lx} = L_x \frac{di_{Lx}}{dt}$$

$$\frac{\Delta i_{Lx}}{\Delta t} = \frac{\Delta i_{Lx}}{(1-D)T} = \frac{-V_0}{L_x}$$

$$(\Delta i_{Lx})_{open} = \frac{-V_0(1-D)T}{L_x}$$

7.4 포워드 변환기: 전류

- Switch On

$$(\Delta i_{Lx})_{closed} = \left[V_S\left(\frac{N_2}{N_1}\right) - V_0\right]\frac{DT}{L_x}$$

- Switch Off

$$(\Delta i_{Lx})_{open} = \frac{-V_0(1-D)T}{L_x}$$

- 충전전류+방전전류=0

$$\left[V_S\left(\frac{N_2}{N_1}\right) - V_0\right]\frac{DT}{L_x} - \frac{V_0(1-D)T}{L_x} = 0,$$

$$\left[V_S\left(\frac{N_2}{N_1}\right) - V_0\right]DT = V_0(1-D)T$$

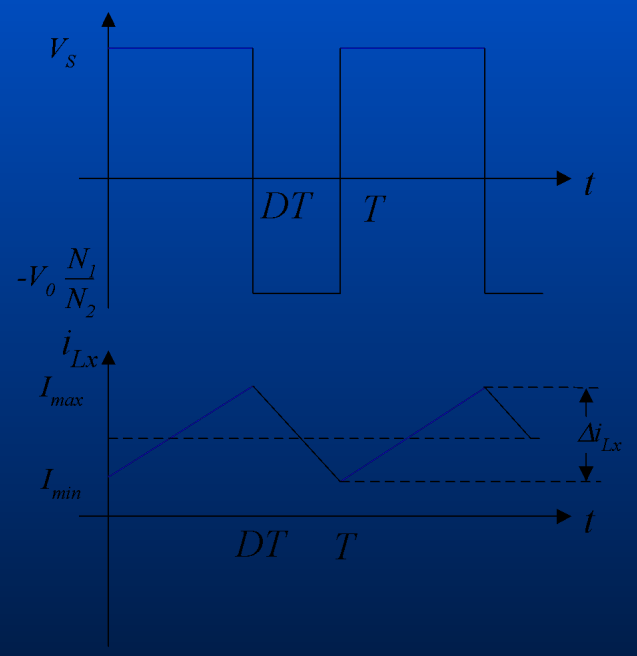
$$V_S\left(\frac{N_2}{N_1}\right)DT - V_0DT = V_0(1-D)T,$$

$$V_S\left(\frac{N_2}{N_1}\right)DT = V_0(1-D)T + V_0DT$$

$$V_S\left(\frac{N_2}{N_1}\right)DT = \frac{V_0}{L_x}((1-D)T + DT)$$

$$\therefore V_0 = V_S D\left(\frac{N_2}{N_1}\right)$$

7.4 포워드 변환기 : 인덕터전류



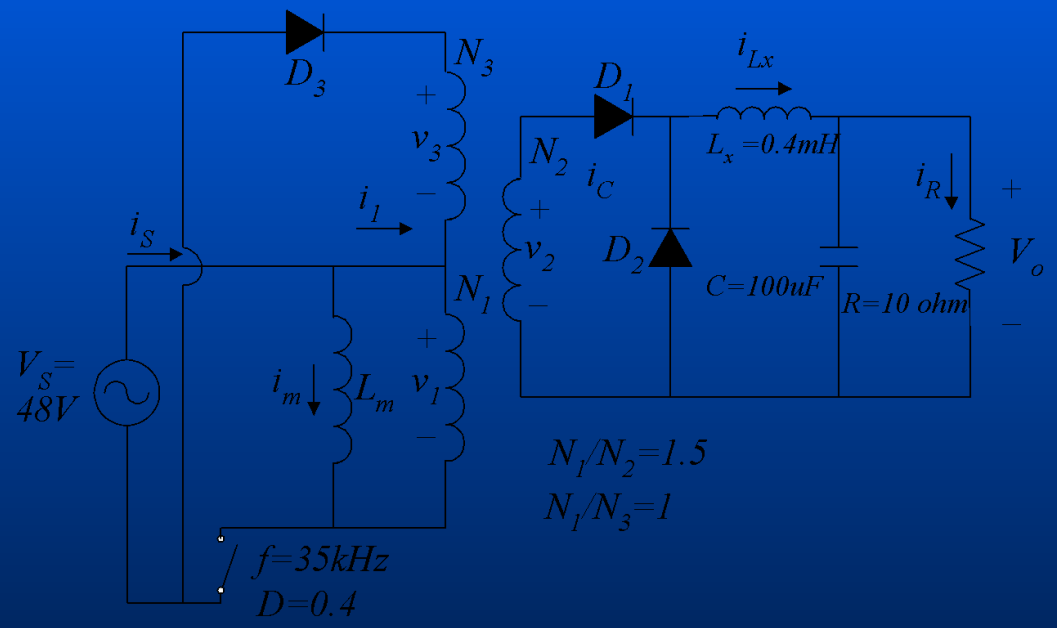
- 인덕터의 맥동전류

$$\frac{\Delta i_{Lx}}{\Delta t} = \frac{\Delta i_{Lx}}{(1-D)T} = \frac{-V_0}{L_x}$$

$$\therefore \Delta i_{Lx} = \left| \frac{-V_0(1-D)T}{L_x} \right|$$

$$= \frac{V_0(1-D)}{L_x f}$$

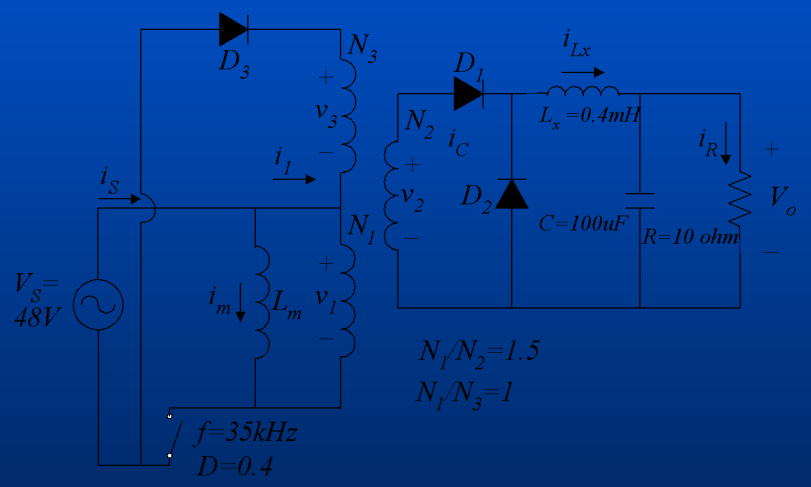
예제 7-4 포워드 변환기(1)



(a)출력전압

$$V_0 = V_S D \frac{N_2}{N_1} = 48(0.4)\left(\frac{1}{1.5}\right)$$

예제 7-4 포워드 변환기(2)



(a)평균전류

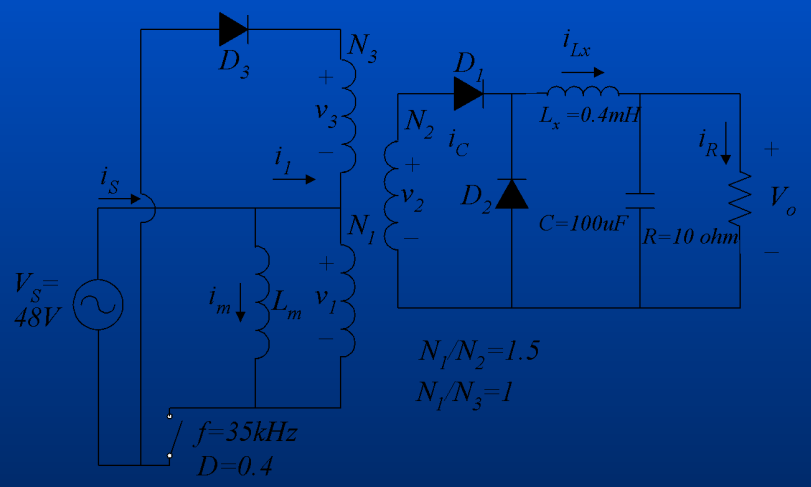
$$I_{Lx} = \frac{V_0}{R} = \frac{12.8}{10} = 12.8A$$

$$\Delta i_{Lm} = \frac{V_S(1-D)}{L_x f} = \frac{(12.8)(1-0.4)}{0.4(10)^{-3}(35000)} = 0.55A$$

$$I_{Lx,\max} = I_{Lx} + \frac{\Delta i_{Lx}}{2} = 1.28 + \frac{0.55}{2} = 1.56A$$

$$I_{Lx,\min} = I_{Lx} - \frac{\Delta i_{Lx}}{2} = 1.28 - \frac{0.55}{2} = 1.01A$$

예제 7-4 포워드 변환기(3)



(b)1차 권선의 첨두전류

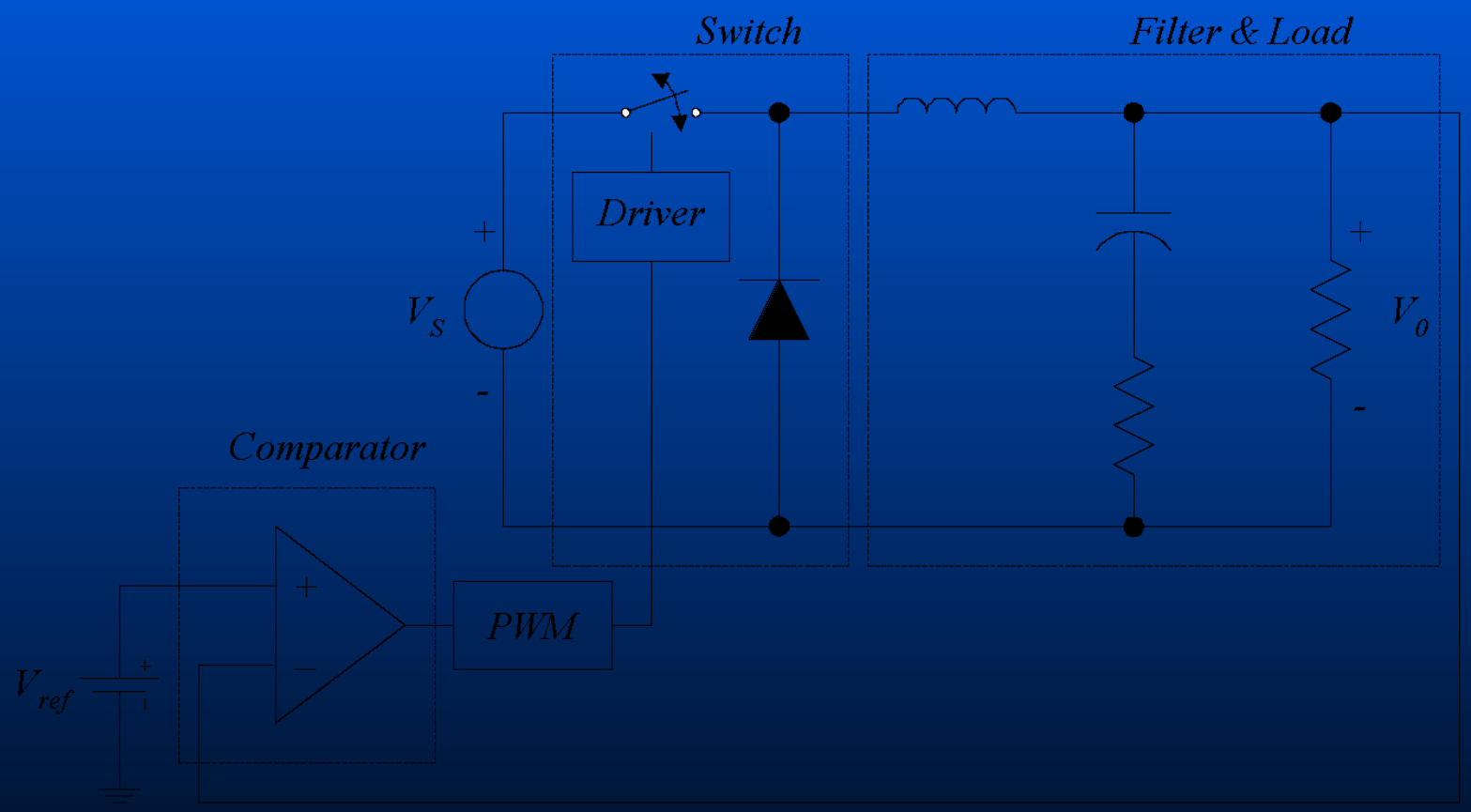
$$I_{Lm,\max} = \Delta I_{Lm} = \frac{V_S DT}{L_m}$$

$$= \frac{48(0.4)}{5(10)^{-3}(35000)} = 0.11A$$

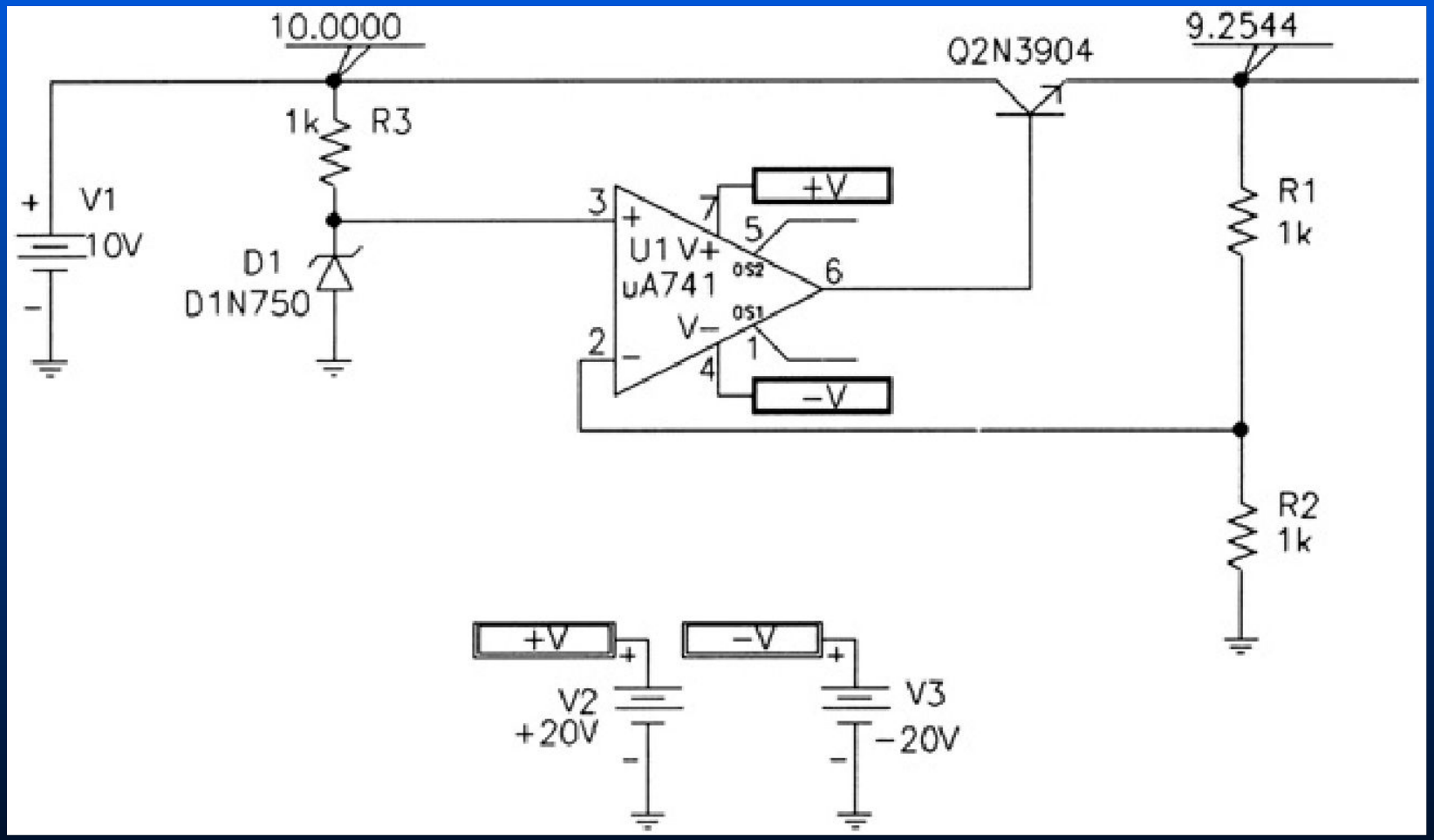
$$I_{\max} = I_{Lx,\max}\left(\frac{N_2}{N_1}\right) + I_{Lm,\max}$$

$$= 1.56\left(\frac{1}{1.5}\right) + 0.11 = 1.15A$$

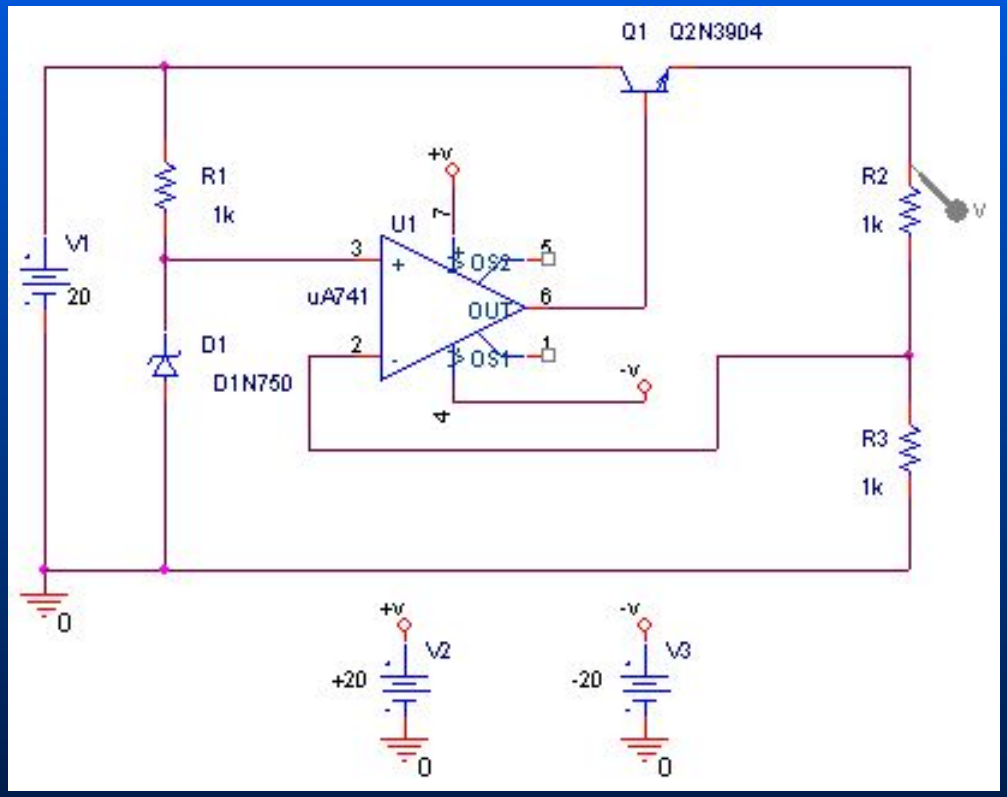
7.12 SMPS(Switched Mode Power Supply)



7.12 SMPS(Switched Mode Power Supply)



예제 7-12 SMPS (OrCAD 해)



- *Parameter*
  - ✔ V_1=20 V
  - ✔ V_{out}=9.25 V
  - ✔ IC : UA 741
  - ✔ V1, R2, R3 변화
  - ✔ Step
    =0 100 ms
- *Find*
  - ✔ 출력전압
  - ✔ 전류 최대, 최소값